House Price Prediction Using Machine Learning Algorithms

Introduction:

Machine Learning is a branch of Artificial Intelligence that helps computers learn from data without being explicitly programmed. It is widely used to solve real-world problems by making predictions and decisions based on patterns in data. Regression is a type of supervised machine learning used to predict continuous values such as house prices, sales, and temperature. In this project, regression algorithms are used to predict house prices using important features like bedrooms, bathrooms, square feet living area, and floors. Linear Regression is the basic algorithm used for prediction. Ridge Regression helps reduce overfitting and improves model performance. Decision Tree Regressor and Random Forest Regressor handle complex data patterns effectively. Support Vector Regressor and KNN Regressor are useful for non-linear predictions. Gradient Boosting Regressor provides high accuracy by improving previous errors step by step. Comparing all these algorithms helps identify the best model for accurate house price prediction.

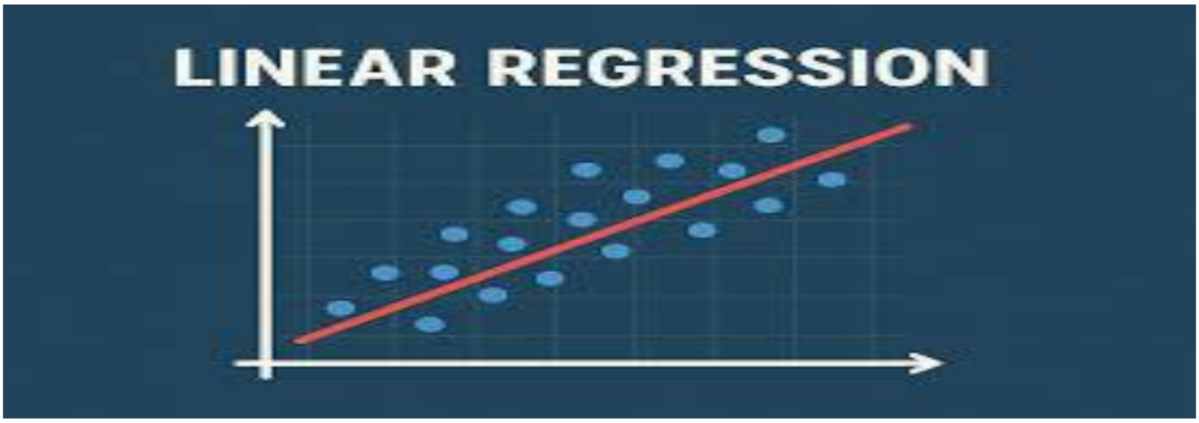
Feature Scaling using Standard Scaler:

Feature Scaling using Standard Scaler is used to bring all input features to the same scale. It helps machine learning algorithms perform better by removing large differences in feature values. Standard Scaler transforms the data so that the mean becomes 0 and the standard deviation becomes 1. This improves the accuracy of algorithms like SVR, KNN, and Linear Regression. It also helps the model train faster and produce better predictions.

Machine Learning Algorithms:

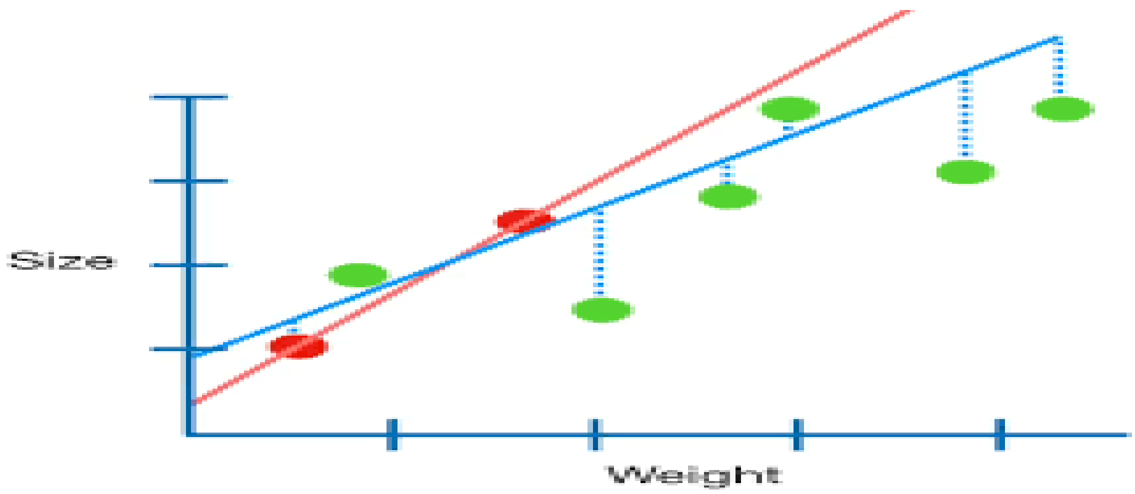
Linear Regression:

Linear Regression is a simple supervised machine learning algorithm used for predicting continuous values like house prices. It finds the relationship between input variables and the target variable using a straight-line equation. The model tries to fit the best line through the data points. It works well when the relationship between variables is linear. It is easy to understand and implement for beginners. It helps in estimating values based on past data. It is widely used in price prediction, sales forecasting, and trend analysis. It gives fast results and requires less computational power.



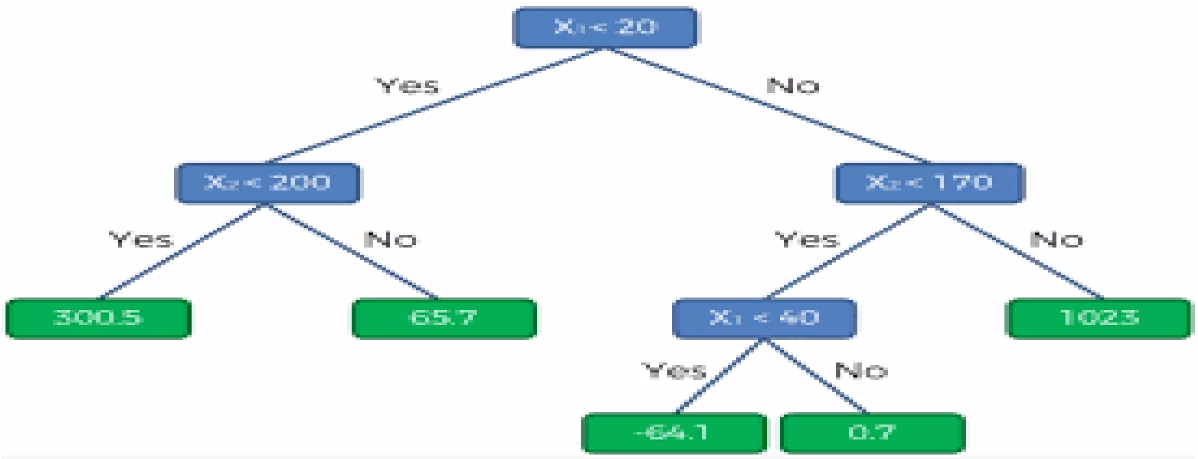
Ridge Regression:

Ridge Regression is an improved version of Linear Regression used to reduce overfitting. It adds a penalty term to the loss function which controls large coefficient values. This helps the model perform better on unseen data. It is useful when the dataset has many correlated features. Ridge Regression improves model stability and accuracy. It is commonly used when normal Linear Regression gives poor performance. It handles multicollinearity problems effectively. It is suitable for large datasets with multiple features.



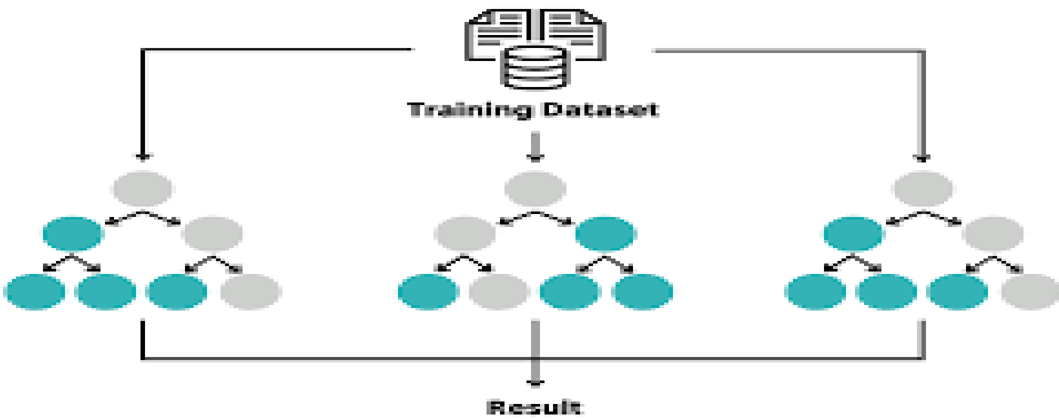
Decision Tree Regressor:

Decision Tree Regressor is used to predict continuous values by splitting data into smaller parts. It works like a tree structure with branches and decision rules. Each split is based on feature values that help improve prediction. It is easy to understand and visualize. It does not require feature scaling like other algorithms. It can handle both small and large datasets. However, it may sometimes overfit the training data. It is commonly used in house price prediction and business forecasting.



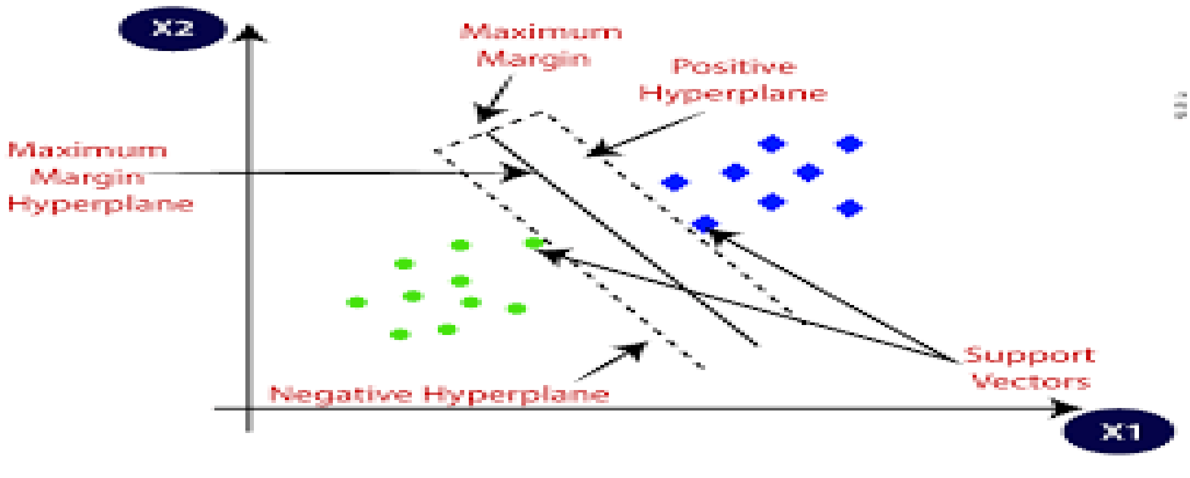
Random Forest Regressor:

Random Forest Regressor is an advanced version of Decision Tree Regressor. It uses multiple decision trees instead of a single tree. Each tree gives a prediction, and the final result is the average of all trees. This improves accuracy and reduces overfitting. It performs well on large and complex datasets. It can handle missing values and non-linear relationships effectively. It is one of the most powerful regression algorithms. It is widely used in real-world prediction problems.



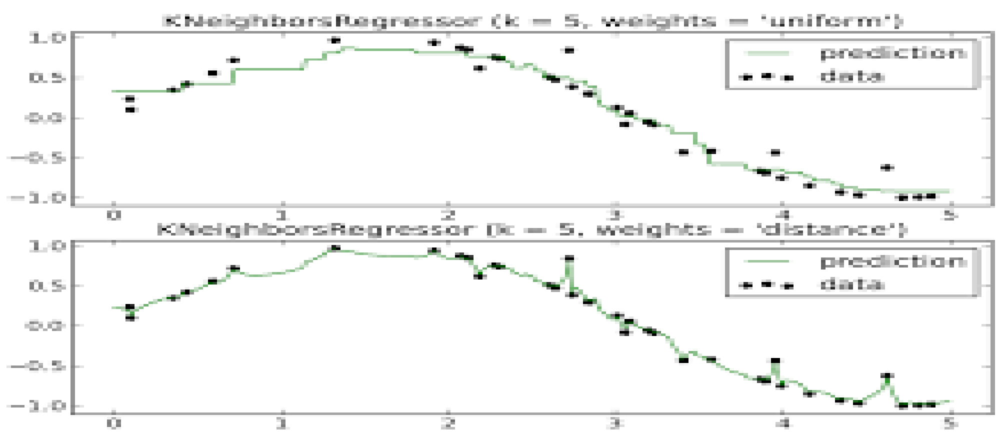
Support Vector Regressor:

Support Vector Regressor is used for predicting continuous values while maintaining a margin of tolerance. It tries to fit the best line within a certain error range. It works well for both linear and non-linear datasets. SVR is effective when the dataset is complex and has many features. It provides high accuracy for difficult prediction tasks. It requires proper parameter tuning for better results. It is commonly used in stock prediction and price forecasting. It performs well with medium-sized datasets.



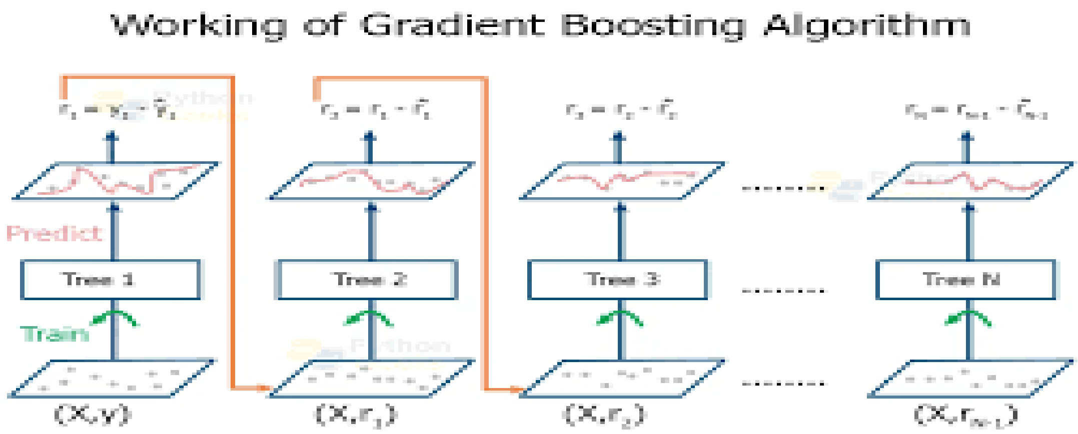
KNN Regressor:

K-Nearest Neighbors Regressor predicts values based on nearby data points. It finds the nearest neighbors of a new data point and calculates the average value. It is simple and easy to understand for beginners. It does not require training like other algorithms. It works well for small datasets with fewer features. However, it becomes slow for large datasets. It is sensitive to irrelevant features and scaling issues. It is commonly used for recommendation systems and prediction tasks.



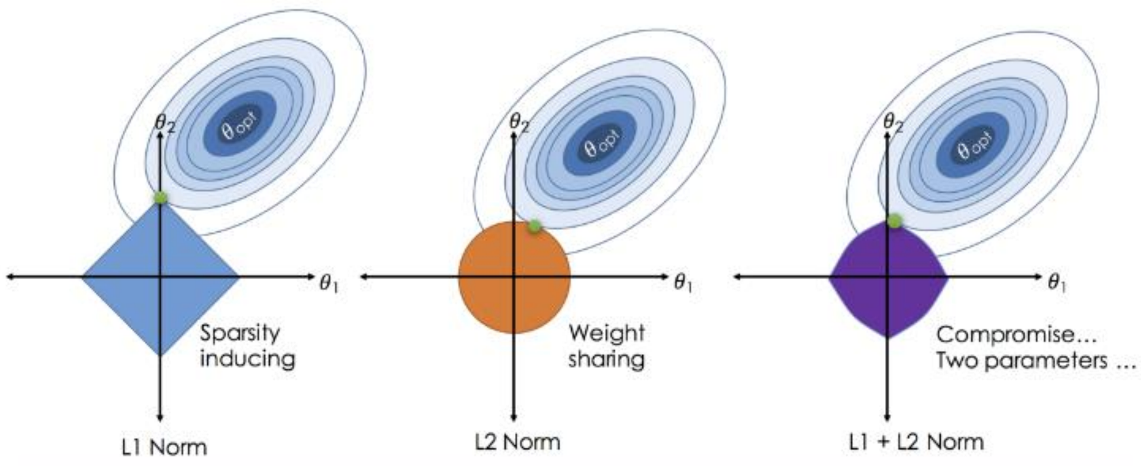
Gradient Boost Regressor:

Gradient Boosting Regressor is a powerful machine learning algorithm used for high-accuracy predictions. It builds models step by step by correcting previous errors. Each new model improves the mistakes made by the earlier model. It combines many weak models to create a strong prediction model. It works very well for complex and large datasets. It provides better performance than many basic regression algorithms. It is widely used in competitions and real-world business problems. It requires more training time but gives excellent results.



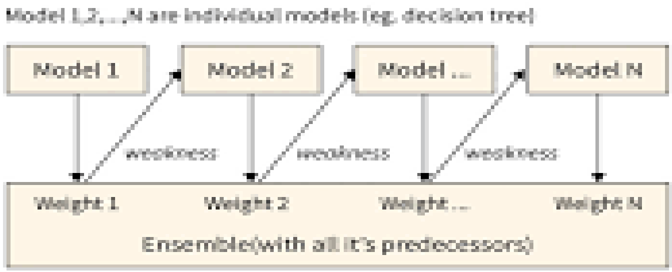
Implementing ElasticNet Regression:

ElasticNet regression is a linear regression model that combines both \(L1\) (Lasso) and \(L2\) (Ridge) regularization penalties, allowing it to handle multicollinearity and perform feature selection simultaneously. Implemented primarily using scikit-learn in Python, the process involves standardizing features, using ElasticNet() with alpha (regularization strength) and l1_ratio (mix ratio), and fitting with .fit().



Implementing AdaBoost Regressor:

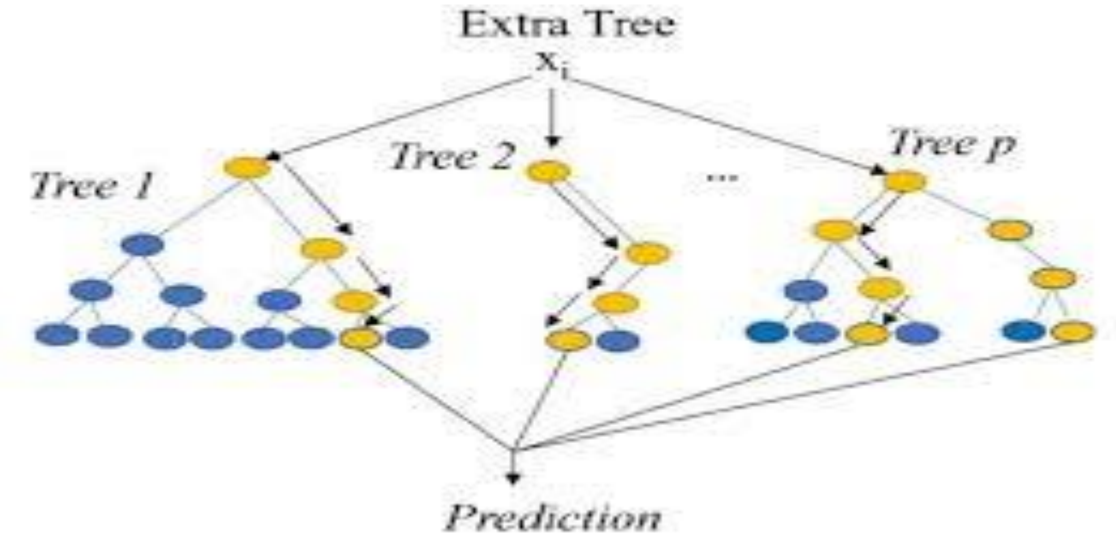
An AdaBoost Regressor (Adaptive Boosting) is an ensemble machine learning algorithm designed for regression tasks, which improves predictive accuracy by sequentially training multiple "weak" learners (typically simple decision trees) to correct the errors of their predecessors. Unlike classification, which focuses on misclassified data points, the AdaBoost regressor adapts by placing higher weights on training samples that yield larger errors, forcing subsequent models to focus on the most difficult-to-predict cases.



Implementing Extra Trees Regressor:

The Extra Trees Regressor (Extremely Randomized Trees) is an ensemble machine learning algorithm that combines the predictions from multiple decision trees to produce a continuous numerical output, often with higher speed and lower variance than Random Forest.

Unlike Random Forest, which searches for the *best* split among a subset of features, Extra Trees picks random split points for each feature, which increases the diversity of trees and reduces overfitting.



Conclusion:

In this project, different regression algorithms were applied to predict house prices using machine learning techniques. Algorithms such as Linear Regression, Ridge Regression, Decision Tree Regressor, Random Forest Regressor, Support Vector Regressor, KNN Regressor, and Gradient Boosting Regressor were used and compared. Feature scaling using Standard Scaler helped improve the performance of the models. Each algorithm has its own advantages depending on the dataset and prediction requirements. Among all models, advanced algorithms like Random Forest and Gradient Boosting usually provide better accuracy for complex data. Comparing all these models helps in selecting the best algorithm for accurate house price prediction. This project shows how machine learning can be effectively used in real-world price prediction problems.

What I Have Learnt?

I learnt the basics of Machine Learning and how regression algorithms are used for house price prediction. I understood how to preprocess data and select important features from the dataset. I learnt the importance of Feature Scaling using Standard Scaler for better model performance. I also understood how to apply different regression algorithms and compare their results. This project improved my practical knowledge of machine learning and prediction models.